

Original Article

Spontaneous regression and rare relapse after excisional biopsy in long-term observation of 31 patients with primary conjunctival lymphoma

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To elucidate long-term outcome in primary conjunctival lymphoma, a review was conducted of 31 consecutive patients: 21 men and 10 women with an age range of 28 to 85 (median, 61) years at presentation and follow-up periods ranging from 1 to 19 (median, 7) years. Conjunctival lymphoma was on the right side in 10 patients, on the left side in 12, and on both sides in 9. Upper, lower, or both fornix lesions in 28 patients were all diagnosed as extranodal marginal zone B-cell lymphoma of mucosa-associated lymphoid tissue (MALT lymphoma), while thick nasal bulbar conjunctival lesions in 3 patients were differently diagnosed as MALT lymphoma, diffuse large B-cell lymphoma, and follicular lymphoma, respectively. Seven patients underwent local radiation (30 Gy): as initial treatment in 5 patients and treatment for relapse in 2 patients. The remaining 24 patients were observed without additional treatment after excisional biopsy: 5 of these 24 patients showed relapse 0.5 to 6 years later and underwent excisional biopsy again that revealed MALT lymphoma. Of the 5 patients with relapse, only one with second-time relapse underwent radiation. Fluorodeoxyglucose positron emission tomography was performed in 18 patients and showed no systemic lesions: high uptake was noted in the residual conjunctival lesions of 4 patients and in the relapsed conjunctival lesions of 3 patients. One patient died of rectal cancer while no patients died of lymphoma. Observation is an option in patients with primary conjunctival lymphoma after excisional biopsy. Radiation is a treatment option in the case of relapse.

Keywords: conjunctival lymphoma, extranodal marginal zone B-cell lymphoma of mucosa-associated lymphoid tissue (MALT lymphoma), follicular lymphoma, diffuse large B-cell lymphoma, fluorodeoxyglucose positron emission tomography (FDG-PET)

INTRODUCTION

The conjunctiva is a part of the ocular adnexa, which are the supporting tissues of the eye ball in the orbit, in addition to other tissues such as lacrimal gland and sac, eyelids, and extraocular muscles. The ocular adnexa are known as one of the dominant sites for lymphoma¹ as well as other inflammatory disorders such as sarcoidosis and IgG4-related disease.²⁻⁴ Lymphomatous lesions of the conjunctiva typically present as salmon-pink lesions and excisional biopsy is required to establish the diagnosis. Primary conjunctival lymphoma appears to be predominant,⁵⁻⁹ while secondary lymphoma as sequelae to conjunctival infiltration with lymphoma at the other sites of the body is relatively rare.¹⁰⁻¹² The predominant pathological type of the primary conjunctival lymphoma is extranodal marginal zone B-cell lymphoma of mucosa-associated lymphoid tissue (MALT lymphoma).

In our previous study published in 2004, we described the long-term follow-up results of observation or radiation in 13 patients with primary conjunctival MALT lymphoma and showed that a large fraction of the patients had spontaneous regression after excisional biopsy.⁵ We advocated at that time that follow-up without additional treatment would be an option for conjunctival MALT lymphoma after pathological diagnosis by excisional biopsy. In this study, we show the follow-up results of 18 additional patients with primary conjunctival lymphoma, together with the 13 previously reported patients, in the era of the staging of lymphoma by fluorodeoxyglucose positron emission tomography (FDG-PET).^{13,14} The main purpose of this study is to report that observation with no additional treatment after excisional biopsy could be an option in primary conjunctival lymphoma.

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METHODS

The medical records of 31 consecutive patients with primary conjunctival lymphoma who were diagnosed and followed at Okayama University Hospital in 32 years from 1992 to 2023 were reviewed. The study was approved as a retrospective study by the ethics committee and conformed to the tenets of the Declaration of Helsinki. Of the 31 patients, the first 13 patients in the 11 years from 1992 to 2002 were reported previously⁵ and additional follow-up was done on these patients. Clinical features, pathological features, and prognosis were analyzed in all 31 patients.

RESULTS

Selected cases are shown in Fig. 1 to Fig. 6. The 31 consecutive patients (Table 1) were 21 men and 10 women with an age at presentation ranging from 28 to 85 (median, 61) years and follow-up periods ranging from 1 to 19 (median, 7) years. Conjunctival lymphoma was on the right side in 10 patients, on the left side in 12, and on both sides in 9. Upper, lower, or both fornix lesions of the conjunctiva in 28 patients were all diagnosed as MALT lymphoma, while nasal bulbar conjunctival thick lesions in 3 patients were differently diagnosed as MALT lymphoma (Case 18, Fig. 2), diffuse large B-cell lymphoma (Case 19, Fig. 3), and follicular lymphoma (Case 30, Fig. 6), respectively. All patients underwent magnetic resonance imaging (MRI) to assess the extension into the orbit, and all showed superficial conjunctival lesions except for one (Case 28) who had extension to the anterior orbit.

Seven patients underwent local radiation (30 Gy in total): as an initial treatment in 5 patients and as treatment for relapse in 2 patients (Cases 10 and 15) which included 1 patient (Case 10) who had undergone 10 Gy radiation after diagnosis of conjunctival lymphoma 22 years prior at another hospital. The reasons for choosing local radiation as a treatment at the initial presentation were as follows: one patient (Case 9) had a history of excisional biopsy of the same location with a diagnosis of MALT lymphoma 5 years prior at a different hospital. Another patient (Case 13) had a hard palate lesion, suspected of lymphoma, one year prior, which regressed spontaneously. A third patient (Case 28) with an upper conjunctival fornix lesion showed a deeper portion of the lesion in the anterior orbit by MRI. The other 2 patients (Cases 7 and 12) simply chose radiation (on both sides in Case 12), based on their wishes. The remaining 24 patients were observed without additional treatment after excisional biopsy: 5 of these 24 patients showed relapse on the unilateral side except for 1 (on both sides in Case 15), and underwent excisional biopsy again 0.5 to 6 years later that revealed MALT lymphoma. Four (Cases 4, 14, 17, and 23) of the 5 patients showed no relapse in the period of observation afterwards ranging from 5 years to 15 years. In contrast, 1 (Case 15, Fig. 1) of those 5 patients underwent local radiation following a second-time relapse 6 years of observation after second-time excisional biopsy. No relapse was noted in all 7

patients with local radiation.

Since the year 2007, whole-body 2-[¹⁸F]fluoro-2-deoxy-D-glucose positron emission tomography combined with computed tomography (FDG-PET/CT) has been available to make the staging of lymphoma after pathological diagnosis. FDG-PET/CT was performed on 18 patients and revealed no systemic lesions, except for in 1 patient (Case 18, Fig. 2) who had retroperitoneal lymph node uptake. This patient (Case 18) was detected to have gastric cancer and underwent gastrectomy. High uptake was noted in the residual conjunctival lesions of 4 patients (Cases 22, 26, 27, 28, Fig. 4, Fig. 5) after the excisional biopsy and also in the relapsed conjunctival lesions of 3 patients (Cases 14, 15, 23). No patients died of lymphoma in the present series of 31 patients. Five patients had a history of cancers, while 1 patient (Case 14) died of rectal cancer at 15 years of follow-up after the second excisional biopsy for conjunctival relapse and the observation.

DISCUSSION

A clinical question in this study was whether observation with no additional treatment would be an option in patients with primary conjunctival lymphoma after pathological diagnosis by excisional biopsy. The conjunctiva plays a crucial role in maintaining the ocular surface and should be left intact as much as possible in cases of malignancy. It is, of course, desirable to resect a localized lesion totally with a well-defined margin of the normal tissue. However, lymphomatous lesions in the ocular adnexa are usually not so well-demarcated, and thus, total resection of the lesion is not guaranteed. In general, it is the standard for diagnosis and treatment of lymphomatous lesions to resect partially as an excisional biopsy to establish the pathological diagnosis as a first step, and then to proceed to treatment, such as chemotherapy and local radiation, as the next step. The conjunctiva is easy to observe at usual clinical settings without special equipment, and relapse is therefore easy to detect. In this context, observation would be a treatment option in shared decision-making with patients.

A typical clinical feature of conjunctival lymphoma is salmon-pink lesions, which are usually located in the upper and lower conjunctival fornix. The pathological diagnosis is predominantly MALT lymphoma. Bilateral MALT lymphoma was frequently noted in the conjunctiva as in the other ocular adnexal lymphomas.¹ In this review of 31 consecutive patients, it should be noted that 2 patients showed other pathological diagnoses of lymphoma, including diffuse large B-cell lymphoma (Case 19, Fig. 3) and follicular lymphoma (Case 30, Fig. 6). These 2 patients presented with a common clinical feature, a salmon-pink thick lesion in the bulbar conjunctiva on the nasal side. In contrast, another patient (Case 18, Fig. 2) with the same clinical feature as the nasal bulbar conjunctival lesion was diagnosed with MALT lymphoma. In these 3 patients with thick conjunctival lesions on the nasal side, the entire lesions were resected together with the normal conjunctival margin since the thick lesions were well localized. Since the thick nasal lesions at clinical presentation

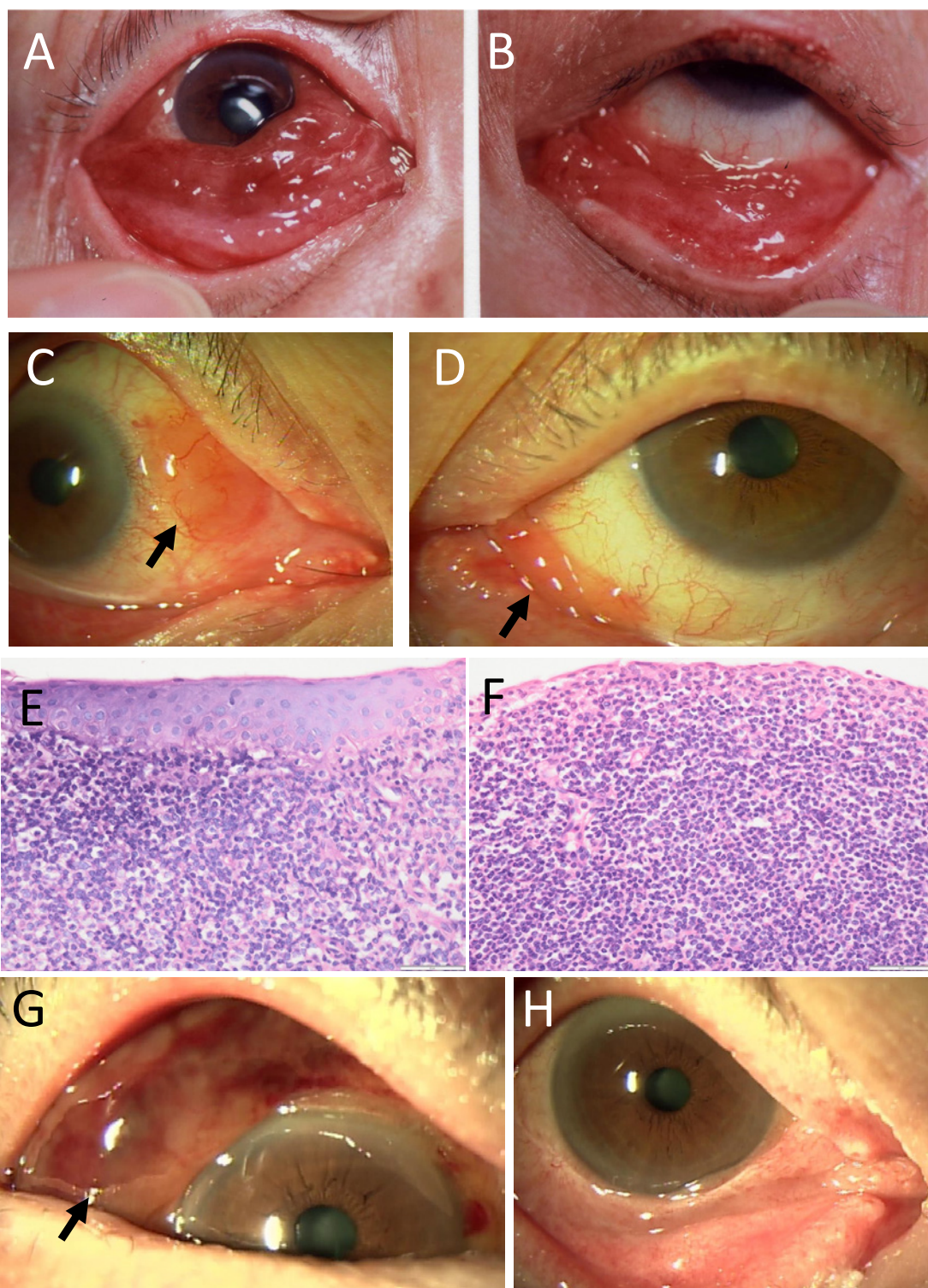


Fig. 1. Case 15. A 72-year-old man. Large lower fornix lesions approaching to the corneal limbus in the right eye (**A**) and left eye (**B**). MALT lymphoma was diagnosed by excisional biopsy in both eyes. Radiation was planned for residual lesions, especially in the right eye, but was postponed because of reactivated tuberculosis. Over two years, the residual lesions in both eyes regressed spontaneously.¹⁵ Relapse as nasal bulbar conjunctival lesions in the right eye (arrow, **C**) and left eye (arrow, **D**) after 4.5 years of observation with no additional treatment after the first excisional biopsy in both eyes. Medium-sized lymphoid cells below the conjunctival epithelium at the second excisional biopsy in the right eye (**E**) and left eye (**F**). Relapse as an upper bulbar conjunctival thick lesion (arrow, **G**) after 6 years of observation after the second excisional biopsy. Diagnosis by third excisional biopsy confirmed MALT lymphoma again. No lesion in the right eye (**H**) 1 year after local radiation (30 Gy) which the patient chose at second relapse. No relapse was noted in 8 years after the radiation. Scale bar = 50 μ m in **E** and **F**.

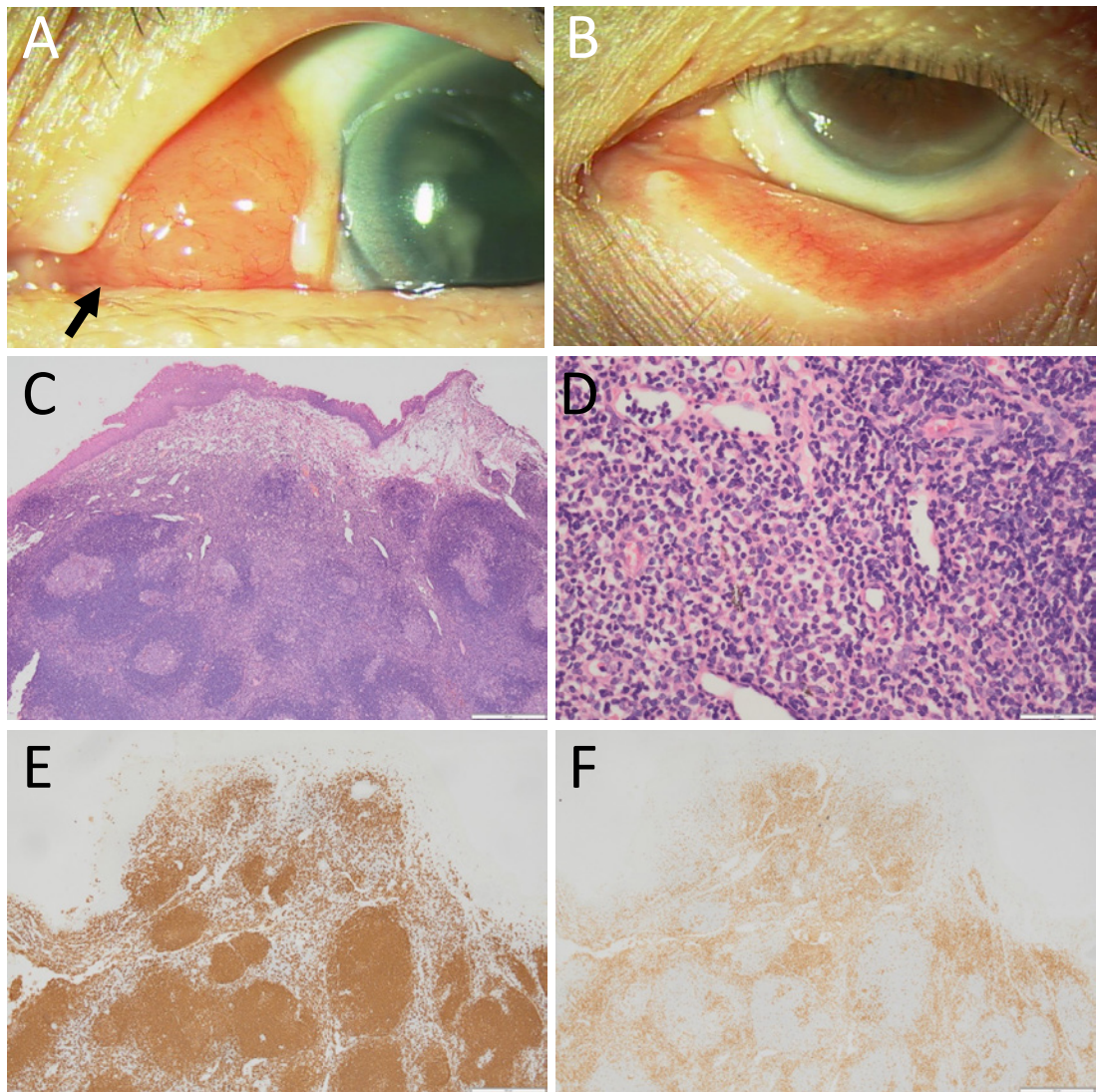


Fig. 2. Case 18. An 80-year-old woman. Nasal bulbar conjunctival thick lesion in the left eye (arrow, **A**). No lesion in the left eye 2 months after excision (**B**). Medium-sized lymphoid cells (**D**), positive for CD20 (**E**), in a diffuse pattern and also around deformed follicles (**C**). A small number of cells positive for CD3 (**F**). The diagnosis was MALT lymphoma with remnant mucosa-associated lymphoid follicles. Fluorodeoxyglucose positron emission tomography for the staging of lymphoma revealed retroperitoneal lymph node uptake, leading to the detection of gastric cancer. The patient underwent gastrectomy. No relapse of lymphoma was noted in 6 years of observation. Scale bar = 500 μ m in **C**, **E**, and **F**. Scale bar = 50 μ m in **D**.

were different from fornix lesions with a common manifestation of conjunctival MALT lymphoma, different diagnostic entities of lymphoma were anticipated at surgery. The patient (Case 19) with a diagnosis of diffuse large B-cell lymphoma was satisfied with the complete removal of the conjunctival lesion and did not want additional treatment because of her old age of 85 years.

In the case of relapse, repeat pathological diagnosis by excisional biopsy is recommended since the relapsed lesion would have different pathological entities of lymphoma.^{16–19} In the present 24 patients who chose observation after the diagnosis of MALT lymphoma by excisional biopsy, 5 patients showed local relapse and the second excisional biopsy showed the same pathological diagnosis. These 5 patients chose observation again and showed no relapse after-

wards, except for 1 patient (Case 15, Fig. 1) who had a second-time relapse and underwent radiation. Excisional biopsy was repeated in these patients with relapse because the conjunctival lesions could be easily approached using local anesthesia. Radiation or intravenous rituximab monotherapy would be a treatment option on the occasion of relapse, but patients still chose observation. Radiation was recommended strongly in only 1 patient (Case 15, Fig. 1) who showed second-time relapse.

Fluorodeoxyglucose positron emission tomography is currently recognized as the standard procedure for the staging of lymphoma. In this study, fluorodeoxyglucose positron emission tomography was performed after the pathological diagnosis of lymphoma by excisional biopsy since the year 2007 when the examination became available. In some

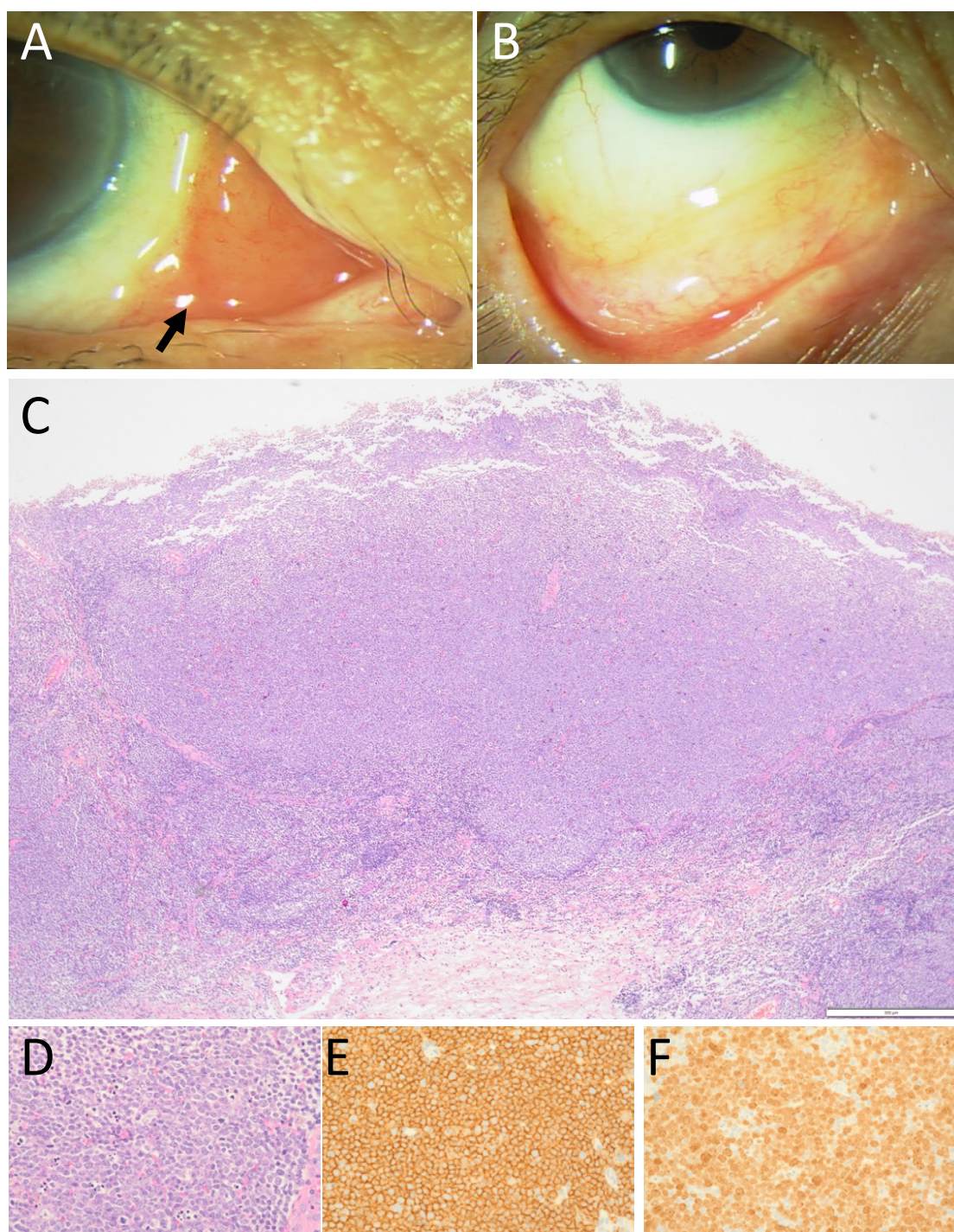


Fig. 3. Case 19. An 85-year-old woman. Nasal bulbar conjunctival thick lesion in the right eye (arrow, **A**). No lesion in the right eye 3 months after excision (**B**). Relatively large lymphoid cells (**D**), positive for CD20 (**E**) and Ki-67 (**F**), in a diffuse pattern of infiltration (**C**), were diagnosed as diffuse large B-cell lymphoma. Fluorodeoxyglucose positron emission tomography showed no abnormal uptake systemically. The patient wished for no additional treatment and no relapse was noted in 1 year. Scale bar = 500 μ m in C, 50 μ m in D, E, and F.

patients, follow-up positron emission tomography was done when the residual lesions remained large enough to be noted clinically (Case 26, Fig. 4). In regard to lymphoma, no systemically abnormal uptake sites were found. In the process of the staging, one patient (Case 18, Fig. 2) was found by chance to have gastric cancer and underwent gastrectomy. It should be noted that relapsed lesions and residual lesions

could be detected by fluorodeoxyglucose positron emission tomography.^{13,14}

In conclusions, typical salmon-pink lesions in the upper and lower fornix and both fornices of the conjunctiva are diagnosed pathologically as MALT lymphoma. The thick bulbar conjunctival lesions on the nasal side are rare and diagnosed as other pathological types of lymphoma such as

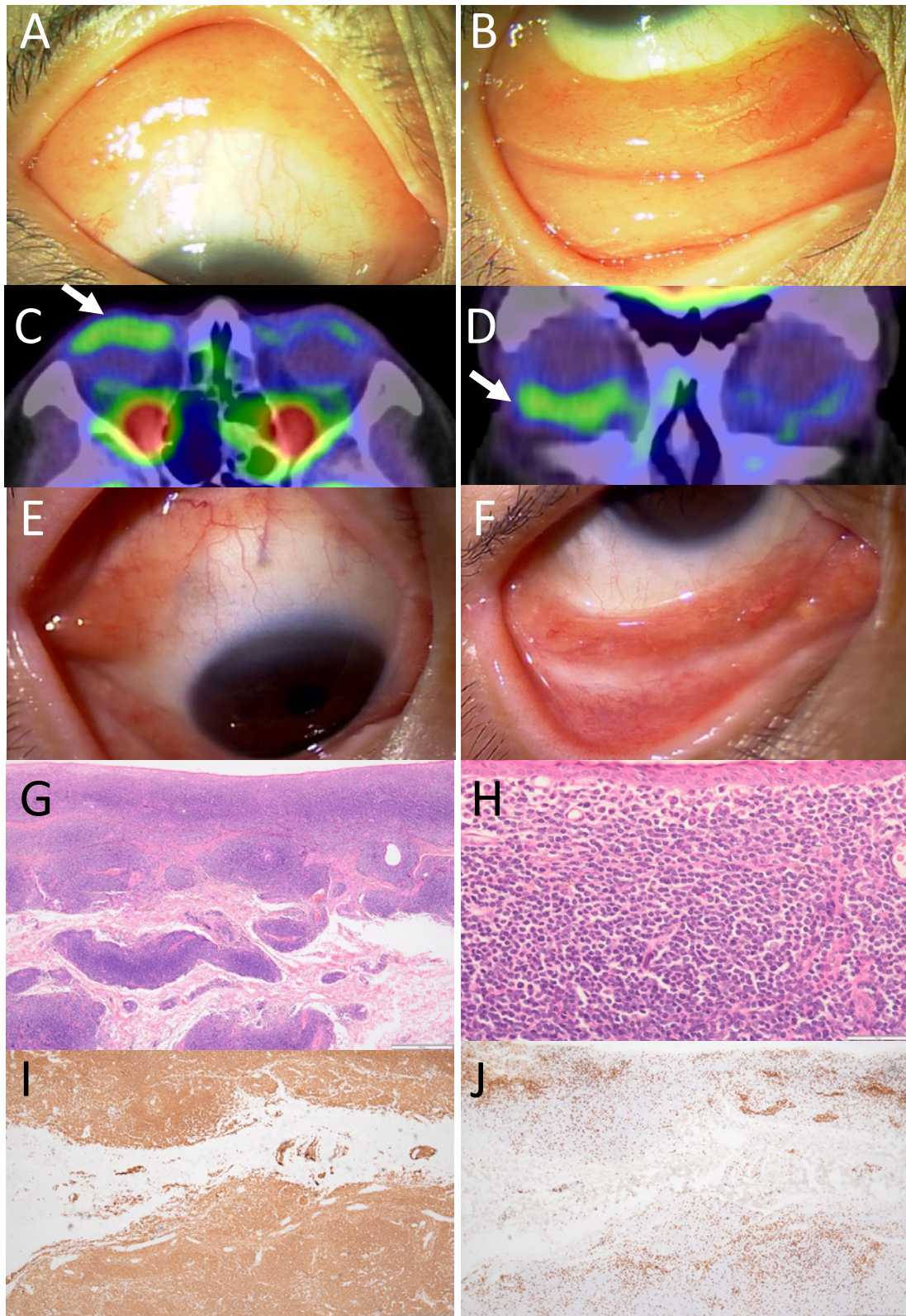


Fig. 4. Case 26. A 56-year-old woman. Upper fornix lesion (**A**) and lower fornix lesion (**B**) in the right eye. Uptake in upper conjunctiva (arrow, **C**) and lower conjunctiva (arrow, **D**) in the right eye by fluorodeoxyglucose positron emission tomography 1 year after excisional biopsy. No abnormal uptake was noted systemically. The patient showed a residual conjunctival lesion in the right eye, but chose observation. The upper (**E**) and lower (**F**) conjunctiva in the right eye appears clear 11 years later at observation. Medium-sized lymphoid cells (**H**), positive for CD20 (**I**), in a diffuse pattern and also around deformed follicles (**G**). A small number of cells were positive for CD3 (**J**). The diagnosis was MALT lymphoma with remnant mucosa-associated lymphoid follicles. Scale bar = 500 μ m in **G**, **I**, and **J**. Scale bar = 50 μ m in **H**.

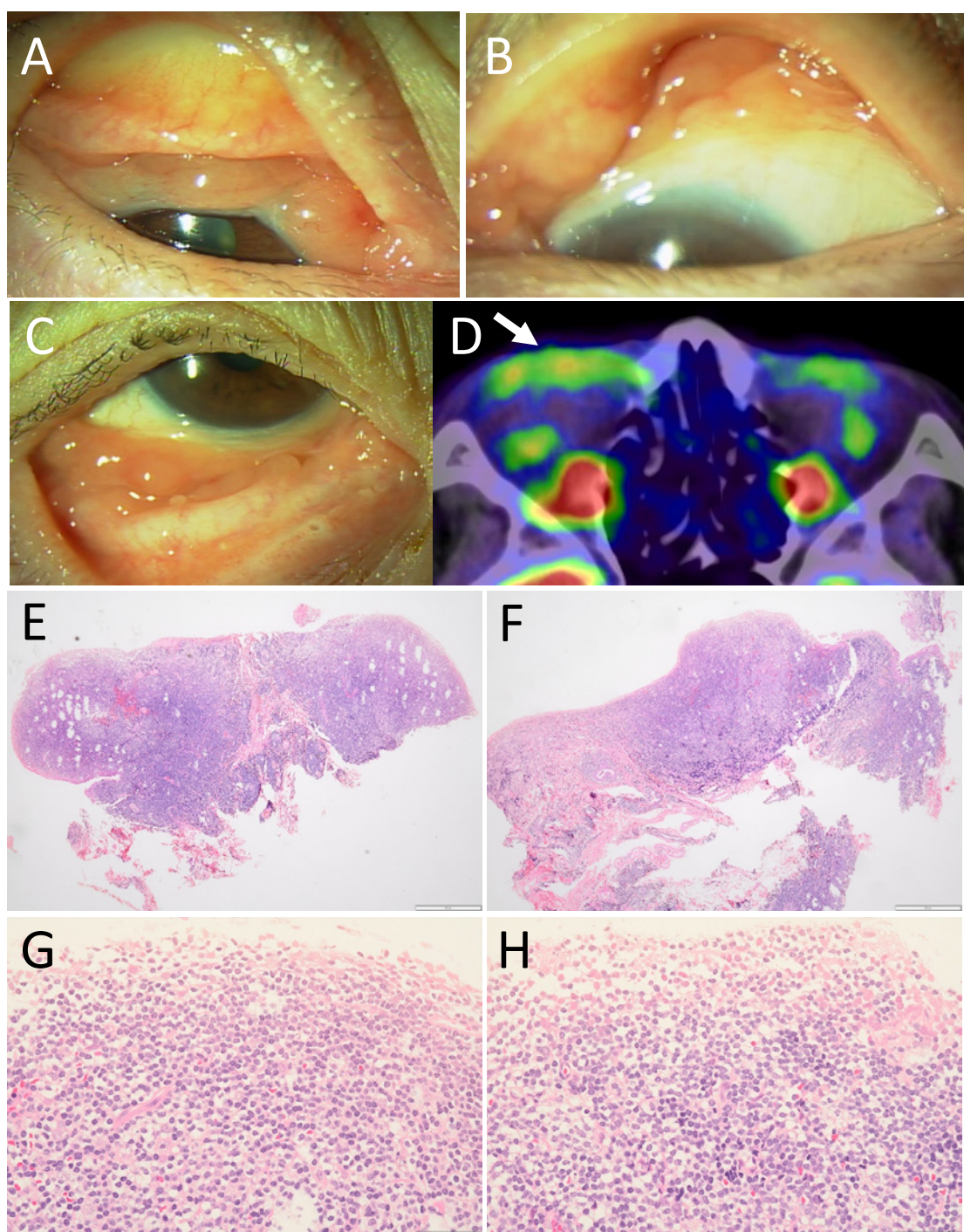


Fig. 5. Case 27. An 83-year-old woman. Upper conjunctival fornix lesions in the right eye (*A*) and left eye (*B*), and a lower conjunctival fornix lesion in the right eye (*C*). Conjunctival uptake, corresponding to bumpy lesions of the upper fornix, in fluorodeoxyglucose positron emission tomography (arrow, *D*). Medium-sized lymphoid cells below the conjunctival epithelium in the right eye (*E*, *G*) and left eye (*F*, *H*). The diagnosis was MALT lymphoma. The patient showed no relapse in 1 year of observation. Scale bar = 500 μ m in *E* and *F*. Scale bar = 50 μ m in *G* and *H*.

diffuse large B-cell lymphoma and follicular lymphoma. Observation with no additional treatment is a recommended option for primary conjunctival lymphoma after excisional biopsy even though the residual lesion may be present. The relapsed lesions should be diagnosed again pathologically by second excisional biopsy and may be followed without addi-

tional treatment. Local radiation is a treatment option in the case of relapse after excisional biopsy when the residual lesion appears large in size.

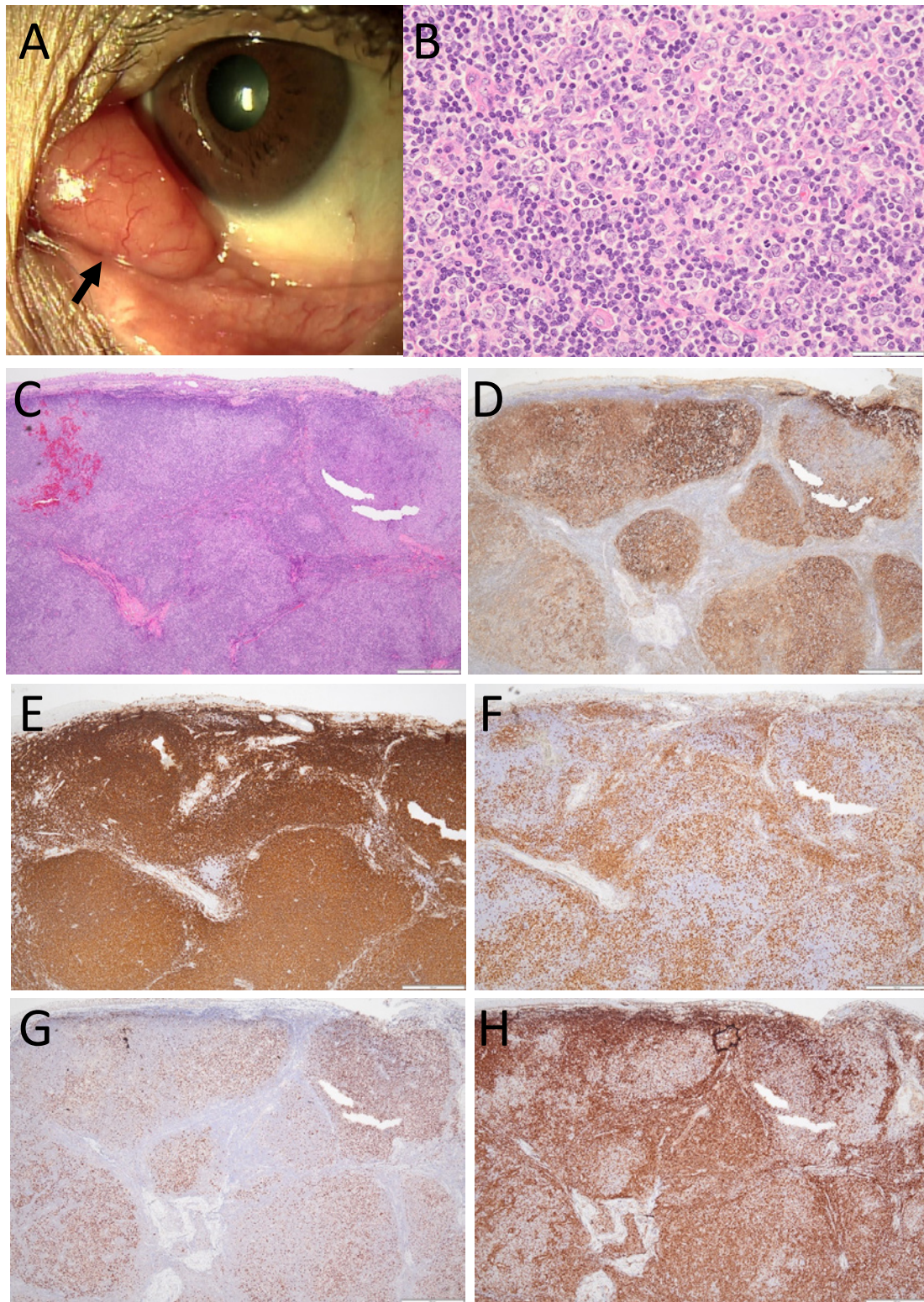


Fig. 6. Case 30. A 68-year-old woman. A nasal bulbar conjunctival thick lesion in the left eye (arrow, **A**). Medium-sized lymphoid cells (**B**) in follicular structures (**C**). Follicular structures with a wide area of the follicular center are formed conspicuously by CD21-positive follicular dendritic cells (**D**) and predominant CD20-positive cells (**E**), admixed with a small number of CD3-positive cells (**F**). Lymphoid cells were positive for bcl-6 (**G**) and negative for bcl-2 (**H**) in a mirror image, indicative of follicular lymphoma. The patient showed no relapse in 2 years of observation. Scale bar = 50 μ m in **B**. Scale bar = 500 μ m in **C–H**.

Table 1. Spontaneous regression, relapse, and radiation in 31 patients with primary conjunctival lymphoma

Case No. / Gender	Age at initial visit (years)	Year of biopsy	Follow-up period (years)	Laterality	Conjunctival location	Pathological diagnosis	Spontaneous regression (year)	Relapse (year)	Year of second resection and diagnosis	Radiation (year)	FDG-PET/CT	Other Features
1/Female	64	1992	15	Left	Upper fornix	MALT	Yes (1996)	No		No	Not done	History of uterus cancer
2/Female	32	1994	19	Right	Upper and lower fornix	MALT	Yes (1997)	No		No	No uptake (2007)	History of allergic conjunctivitis
3/Female	56	1994	9	Bilateral	Right lower fornix Left upper and lower fornix	MALT MALT	Yes (1996)	No		No	Not done	
4/Male	68	1995	18	Left	Upper fornix to limbus	MALT	Yes (1998) after second resection	Yes	1998, Left MALT	No	No uptake (2008)	History of stomach and prostate cancer, 60 Gy radiation for laryngeal cancer in 2009
5/Male	53	1997	6	Bilateral	Right upper and lower fornix Left upper fornix	MALT Not done	Yes (2002)	No		No	Not done	
6/Female	59	1998	5	Left	Upper and lower fornix	MALT	Yes (1999)	No		No	Not done	Multisystem atrophy
7/Female	61	1999	14	Left	Upper and lower fornix	MALT	Not applicable	No		30 Gy (1999)	Not done	History of thyroid cancer
8/Male	66	1999	4	Left	Upper fornix	MALT	Yes (2002)	No		No	Not done	
9/Male	39	1999	4	Left	Upper and lower fornix	MALT	Not applicable	No		30 Gy (1999)	Not done	Previous biopsy of the same location (1994); MALT lymphoma
10/Female	63	1999	5	Right	Upper and lower fornix	MALT	Not applicable	No		30 Gy (2002)	Not done	Previous biopsy of the same location (1977); lymphoma, 10 Gy radiation (1977)
11/Male	67	2000	7	Right	Lower fornix	MALT	Yes (2002)	No		No	Not done	
12/Female	58	2001	2	Bilateral	Right lower fornix to limbus Left lower fornix	MALT Not done	Not applicable	No		30 Gy (2001) in bilateral	Not done	
13/Male	55	2002	1	Right	Lower fornix	MALT	Not applicable	No		30 Gy (2002)	Not done	Hard palate lesion (2001); spontaneous regression
14/Female	52	2003	19	Right	Upper and lower fornix	MALT	Yes (2007), after second resection	Yes (2007)	2007, Right MALT	No	Right conjunctival uptake in 2007 (SUVmax=2.59) No other uptake	Died of rectal cancer
15/Male	72	2004	19	Bilateral	Right lower fornix to limbus Left lower fornix to limbus	MALT MALT	No	Yes (2009, 2016)	2009, Bilateral MALT 2016, Right MALT	30 Gy, Right (2016)	No uptake in 2007 Bilateral conjunctival uptake in 2009 (SUVmax=2.8) Right conjunctival uptake in 2015 (SUVmax=4.23)	Therapy for recurrent tuberculosis in 2005 Parkinsonism
16/Female	61	2004	19	Left	Lower fornix	MALT	Yes (2005)	No		No	No uptake in 2007 No uptake in 2017	Hypothyroidism
17/Female	66	2004	11	Bilateral	Right lower fornix Left lower fornix	MALT MALT	Yes (2009), after second resection	Yes (2009)	2009, Right MALT	No	No uptake in 2007 No uptake in 2009	

18/Female	80	2009	6	Left	Nasal bulbar	MALT	Yes (2009)	No	No	Retropertoneal lymph node uptake (gastric cancer) in 2009 (SUVmax=6.97)	History of left breast cancer Gastrectomy for gastric cancer in 2011
19/Female	85	2009	1	Right	Nasal bulbar	DLBCL	Yes (2009)	No	No	No uptake in 2009	
20/Female	74	2009	11	Bilateral	Right lower fornix Left lower fornix	MALT	Yes (2010)	No	No	No uptake in 2009 No uptake in 2010 No uptake in 2011	
21/Male	59	2009	7	Right	Lower fornix	MALT	Yes (2009)	No	No	No uptake in 2009 No uptake in 2010	
22/Male	28	2010	6	Right	Upper and lower fornix	MALT	Yes (2011)	No	No	Right conjunctival uptake in 2010 (SUVmax=2.56) No other uptake	
23/Female	55	2010	13	Bilateral	Right lower fornix Left upper fornix to nasal limbus	MALT	Yes (2010), after second resection	Yes (2010) (2010)	No	No uptake in 2010 Left conjunctival uptake half a year later in 2010 (SUVmx=5.18) No other uptake	
24/Female	56	2010	13	Left	Upper fornix	MALT	Yes (2011)	No	No	No uptake in 2010 No uptake in 2011 No uptake in 2015	
25/Female	54	2010	13	Left	Lower fornix	MALT	Yes (2011)	No	No	No uptake in 2011	
26/Female	56	2012	11	Right	Upper and lower fornix	MALT	Yes (2023)	No	No	Right conjunctival uptake in 2012 (SUVmax=2.91) Right conjunctival uptake in 2013 (SUVmax=3.34) No other uptake	
27/Female	83	2012	1	Bilateral	Right upper and lower fornix Left upper and lower fornix	MALT	Yes (2012)	No	No	Bilateral conjunctival uptake in 2012 (SUVmax=3.28/2.52 in right/left) No other uptake	Pulmonary nontuberculous mycobacteriosis
28/Female	80	2013	4	Right	Upper fornix to anterior orbit	MALT	Not applicable	No	30 Gy (2013)	Right conjunctival uptake in 2013 (SUVmax=3.54) Nasal cavity soft tissue uptake (SUVmax=5.04) No uptake in 2015	
29/Male	39	2015	7	Bilateral	Right upper and lower fornix Left upper and lower fornix	MALT	Yes (2015)	No	No		
30/Female	68	2021	2	Left	Nasal bulbar	follicular lymphoma	Yes (2021)	No	No	Not done based on patient's wish	
31/Female	84	2021	1	Left	Lower fornix	MALT	Yes (2021)	No	No	Not done based on patient's wish	Right lower lobectomy for lung cancer in 2021

FDG-PET/CT, fluorodeoxyglucose positron emission tomography/computed tomography; MALT, marginal zone B-cell lymphoma of mucosa-associated lymphoid tissue (MALT); DLBCL, diffuse large B-cell lymphoma.

AUTHORS' NOTE

Data are available upon reasonable request to the corresponding author.

AUTHORS' CONTRIBUTIONS

T.M., as an ophthalmologist, performed surgery, followed the patient, and wrote the manuscript. T.T., as a pathologist, made pathological diagnoses. All authors approved the final version of the manuscript.

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None.

CONFLICT OF INTEREST

The authors declare no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

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ETHICS APPROVAL

The study was approved as a retrospective study by the ethics committee (2311-028, October 6, 2023).

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